

Human-computer interaction-based robotic design and development and control system testing

—Prototyping and implementation testing study of a robotic hand

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Abstract—One of the most challenging issues in the development of existing robotic hand design and development based on existing policies and in everyday people's needs for interactive technologies that can assist patients who have lost hand manipulation due to neurological or stroke induced. This paper focuses on robotic hand designs aimed at assisting post-stroke patients with enhanced human-computer interaction, including four innovative research direction areas. a) Aesthetically, the four-piece human-computer interaction-based robotic hand design (i.e., internal mechanics, five elastic bands, electrodes, and soft skin-friendly material) looks more like sports equipment than a wearable machine. b) Inspired by human palm design engineering, the The human-computer assisted manipulator is an innovative form of mixing soft and hard components and materials, constructed from layers closest to the body made of soft, stretchable fabrics (i.e., neoprene and fly stitch fabric). 3D printed flexible TPU components, which use BOA dials to increase or decrease finger tension. c) For added strength, the product is worn on the palm of the hand, and then tightening the wrist strap prevents unwanted movement This is an innovative patent and is integrated into the glove. This is innovative and patented and the BOA dial integrated in the glove is attached to five elastic bands, one for each finger, which are secured in place by a channel placed on the TPU piece. This makes it lighter in weight. d) Intelligent. By being equipped with sensors, it can be intelligently sensed internally to properly release tension with promoting blood circulation in the wearer's hand, they push the BOA dial in and rotate it to increase tension, and to release tension, the user simply pulls the BOA dial upward. This will help the muscles relax and restore normal fluid movement. The data can be viewed in the accompanying app. After researching and investigating the existing human-machine interaction based robotic hand mostly made of steel, the sensor cannot sense the material of the object well after wearing, temperature, and the set finger bending angle cannot fit the object well. Through experimental data to test the best working area and experimental daily use force, we study and adjust the best for the human-machine interaction-based robotic three-dimensional work treatment area. This is one of the most innovative solutions with the future of the industry, this research focuses on the process of human-computer interaction for the patient's hand is also required for a large number of interaction resources also with human resources. The development of a more skin-friendly, safer, and less expensive human-computer interaction-based manipulator (HES) that can operate by itself is sufficient to support all daily living activities (ADLs) of the hand. Finally, the physical field study experiments also showed that the device is flexible and can be customized by anyone for the right size and use.

Keywords—Human-computer interaction, robotic arm design, control system testing, rehabilitation

I. INTRODUCTION

According to Chinese research centers, the incidence of stroke is increasing at an annual rate of 8.7% and is one of the leading causes of death and disability in adults, while 11.42 million people over 40 years of age in China suffer from stroke [1]. It is estimated that in the future [2], if the patient's nervous system is not maintained in a timely manner, it will impair the normal use of the patient's hand function [3]. It has also been demonstrated that robotic physical assisted therapy for human-computer interaction of the patient's hand after stroke has been well proven effective [4-7]. Also for patients with hand spasticity human-machine interaction processes are human resource intensive [8]. Since such manipulators [9] are used during rehabilitation, including maintenance can also be performed at a nearby clinic. On the contrary, portability is particularly important for up to 66% of patients up to 24 weeks [10] so based on the above research survey, our researchers wanted to develop a develop a more skin-friendly, safer, and less expensive to be able to human-computer interaction-based robotic hand system (HES) that can enhance the grip strength of the intake [11] and can automatically assist [12] enough to support the hand in daily activities of daily living (ADLs), while greatly reducing the pressure on medical and human resources.

II. METHOD

The human-computer interaction robotic device design study is broken down into six real-life device study demonstrations to actively assist users in making it easier to use and operate this product. Figure 1 below shows. The internal skin-friendly material consists of four main layers, mainly divided into the outermost acrylic butadiene styrene (ABS) exoskeleton, the rubber tube, the bottom elastic fabric and the high elastic fabric (Figure 1A). The BOA dial integrated in the glove is connected to five elastic bands, one for each finger, which are secured in place by channels placed on the TPU sheet. This results in a lighter weight (Figure 1B). Based on the research and study of the kinematics, we performed a hand kinematic analysis, by means of Rhino 6.0 model modeling, and studied the multi-body approach (Newtonian and Lagrangian approaches [16].) By simulating the maximum hand activity angle, the activity angles were shown separately through the

front view and top view angles, where the thumb activity angle in the top view was 0-75° and the other hand activity angles were in accordance with the normal grip angle (Figure 1C). The composition of the system is shown from top to bottom mainly consists of a 16mhz processor, circuit board, sensors, muscle sensors, acrylic butadiene styrene (ABS) exoskeleton complex, bottom elastic fabric and high elastic fabric (Figure 1D). Through the overall display

diagram of the hand in real and human body, the most important power units are the main motorized controller of the palm [17], [18-20], [21] and the rotational device that controls each finger (Figure 1E). Considering that the wearing fingers should have comfortable material and safety, soft material was considered as a researchable choice (Figure 1F).

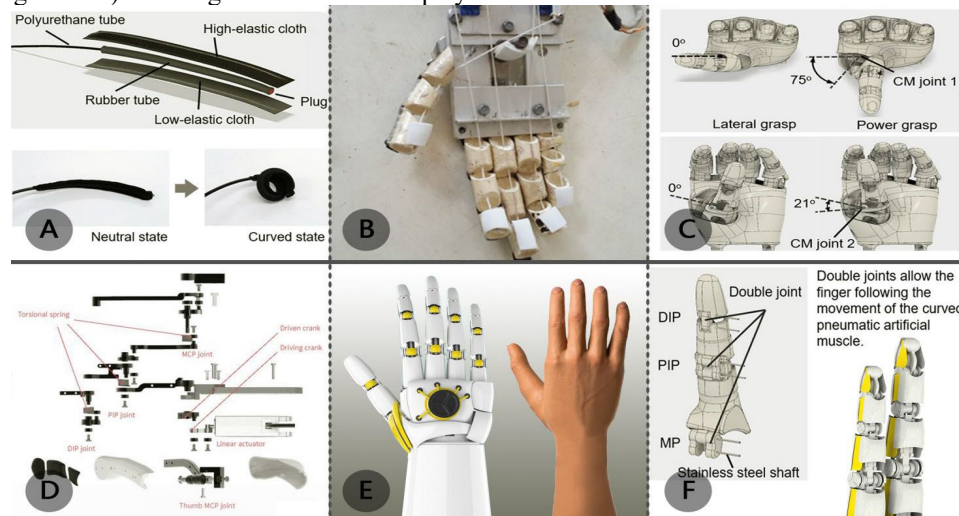


Figure 1. Human-computer interaction-based robotic device design study decomposed into six real device study presentation diagrams. (A) Internal skin-friendly material construction. (B) BOA dial integrated in the glove attached to five elastic bands. (C) Diagram of maximum watch movement angle by simulation. (D) From top to bottom, the system consists of a 16mhz processor, circuit board, sensors, muscle sensors, acrylic butadiene styrene (ABS) exoskeleton, bottom elastic fabric and high elastic fabric. (E) The power unit is the main motorized controller for the palm of the hand and the control of each finger rotation device. (F) Soft material for wearing finger comfort material and safety demonstration diagram.

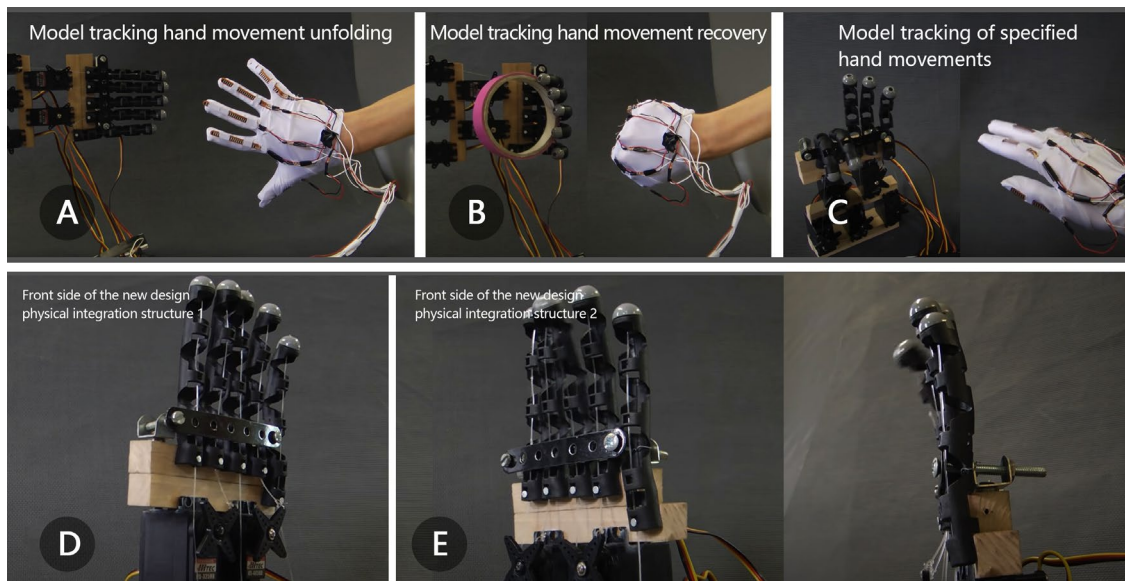


Figure 2. The user operates the human-machine interaction based robotic device by the following three main steps diagram. (A) The hand dynamic sensing detection system will also follow up the data in real time. (B) When we carry out the rehabilitation program to grasp objects, the HCI-based manipulator will also grasp objects from 0.01kg to 1.5kg according to the hand dynamics, and the device enables the patient to better assist the force. (C) Hand commands are used according to the patient's daily use, including handshake, celebration, and special hand command maps. (D) A three-quarter front and side view of the HCI-based manipulator. (E) Three-quarter rear side view of the HCI-based manipulator.

III. MANIPULATOR STRUCTURE FOR HUMAN-COMPUTER INTERACTION

According to the above study human-computer interaction is one of the main design processes of the robotic hand, while we disassemble the model-based training steps, as shown in Figure 2 below, the patient uses the human-computer interaction-based robotic device mainly consists of the following three major steps, firstly, the

patient's hand is put on the soft tissue layer of the new robotic hand we studied, the simulated hand model is fixed, and the hand movement is unfolded, while the model tracks the dynamics of the hand The hand movements are unfolded and the model tracks the hand dynamics to assist in the rehabilitation treatment. At the same time, the induction detection system also follows up the data in real time, including the trajectory of each movement, the strength of each operation and the speed of the grasping operation are

recorded and transmitted in real time (Figure 2A). The human-machine interaction-based manipulator will also grip objects from 0.01 kg to 1.5 kg according to the dynamics of the hand when we carry out the rehabilitation program, and the device enables the patient to better assist the force (Figure 2B). Depending on the operator's daily use, hand commands, including handshakes, celebrations, and special hand commands, can be tracked and demonstrated in real time by the HCI-based manipulator to better assist patients in adapting to daily routines and increase self-confidence (Figure 2C). A three-quarter front and side view of the HCI-based robotic hand shows that the study has obvious advantages due to the entry point of the joints of the palm and fingers. It has high safety, easy to wear, and more ergonomic. In the palm and finger traction also depends on the patient's operational use feedback, through the test joint posture grip angle, grip strength, induction function are more clear (Figure 2D). Based on the human-machine interaction of the robotic hand three-quarter rear side view can be derived, the study needs to consider all kinds of emergency factors, including the emergency locking function, the palm of the twist locking protection function, each finger of the separate locking protection function should be fully considered perfect (Figure 2E).

IV. HUMAN-COMPUTER INTERACTION-BASED PROTOTYPING OF ROBOTIC DEVICES AND GRIPPERS

According to the team's research on the human-computer interaction-based robotic hand prototype with hand drive system, as shown in Figure 3 below, we simulated a realistic kinematic analysis diagram for the physical object combined with the CAD model. Each finger segment is equipped with a drive unit separately. Thus, the assistive forces are transferred to the fingers along the manipulator for the assisted rehabilitation function. For safety reasons, the manipulator is divided into a torque sensor to adjust the traction force and traction angle of each finger separately; a cushioning structure with a DC driver to reduce the friction

of the torque sensor on the skin; a joint rotation axis to better assist the rehabilitation activities of the joint; a soft material to better protect the inner wearer's finger; an acrylic butadiene styrene (ABS) outer bone complex to strengthen the The soft material protects the inner wearer's finger; the ABS outer bone complex strengthens the external structure and steel; the elastic bottom makes it easier and more comfortable for the patient to wear; each joint is equipped with a harness clutch motor to better control each finger joint separately to assist the patient's rehabilitation; the sensors placed at the finger tips can better sense and transmit the corresponding data, including joint posture grip angle, grip strength, and sensing function (Figure 3A). The analysis of the structural diagram in the side view shows that for each finger module there is a drive motor and each finger is designed as a single six-link structure, where each finger consists of three links and the position of the fingertip can be calculated according to the joint structure (a_1, a_2, a_3) (Figure 3B). Recording finger length data and width data of different users, we upgraded the product customization scheme, the structure remains consistent in the joint length for data customization design, providing a new research idea for product development. Our researchers concluded from a 30-day clinical trial with a human-machine interactive robotic hand prototype and hand actuation system that the SG index ($n=24$) criterion for the first week was 53.4 and the CON index ($n=22$) criterion was 48.2 for a sample of 48 males aged 18-56 years old; after four weeks of exercise, the SG index ($n=24$) criterion was 58.3 and the CON index ($n=22$) was 49.6 (Figure 3C). The standard of SG index ($n=24$) for the first week was 32.8; the standard of CON index ($n=22$) was 22.9; the standard of SG index ($n=24$) after four weeks of exercise was 43.1; and the standard of CON index ($n=22$) was 26.4 for the 56 female samples tested, aged 20-62 years (Figure 3D). The tests yielded a significant improvement in both experimental and statistical criteria for the prototype robotic drive system.

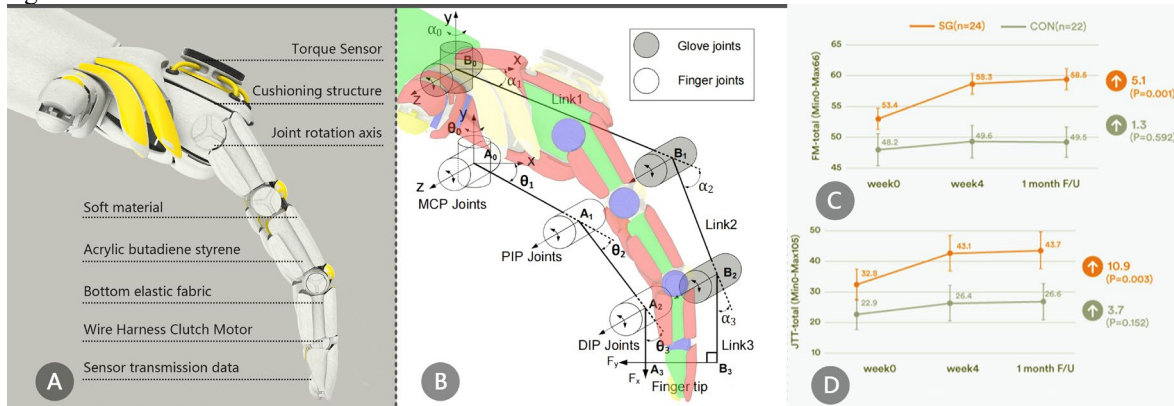


Figure 3. Real kinematic analysis diagram of the real object combined with CAD model for simulation. (A) Diagram of the finger segments of the device structure with separate drive units. (B) Design diagram of a single six-link structure. (C) Structural diagram of a 48-bit male sample that passed the test. (D) Data structure diagram of 56 female samples that passed the test.

V. SIMULATION AND TESTING OF ROBOTIC CIRCUIT DRIVE SYSTEM FOR HUMAN-COMPUTER INTERACTION

The study of the human-machine interaction-based robot circuit drive system simulation and testing, as shown in Figure 4 below, the machine circuit integration design can guarantee daily use. Each mechanical finger has a maximum operating voltage of 16 V. It also has an emergency power-off function and a sensing function. acrylonitrile

butadiene styrene (ABS) can well consider the heat dissipation function during the working hours of the device. the circuit board is perfected in Eagle SoftCad 6.0 development and contains fifteen copies in each PSB. the HCI-based robotic circuit drive system is linked separately component linkage diagram in the sensor system (Figure 4A). The test was conducted by running the main chip of the HCI-based robot circuit drive system in combination with the product motor, which is a square-shaped piece

measuring 52 mm by 54 mm (Figure 4B). A wireframe diagram shows that the main chip is responsible for functional modules including the control of ten individual channel drives; five dual-channel sensing drives with a maximum output torque of 0.536 Nm, which are used to sense the resistance exerted by the fingertip gripping the object, test the torque in real time and transmit the data

individually. Nine motor motors are also the modules responsible for the main chip (Figure 4C). The physical object is worn on the patient's hand and the stability of the product is tested by testing the spacing and fit of the exoskeleton to the hand. The circuitry drives the interaction between the system and the hand for proper bending grip and extension cycles (Figure 4D).

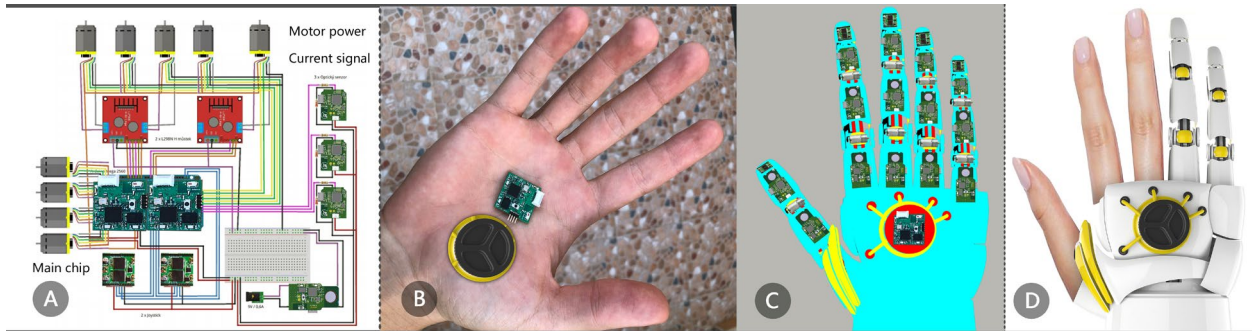


Figure 4. Simulation and test diagram of the robot circuit drive system. (A) Circuit control system drive demonstration diagram. (B) The test is performed by using the main chip of the HCI manipulator circuit drive system in combination with the product motor operation. (C) Wireframe diagram showing the functional modules responsible for the main chip including control of ten individual channel drives; five dual channel induction drives are shown. (D) A diagram of the interaction between the human-machine interactive robotic circuit drive system and the hand.

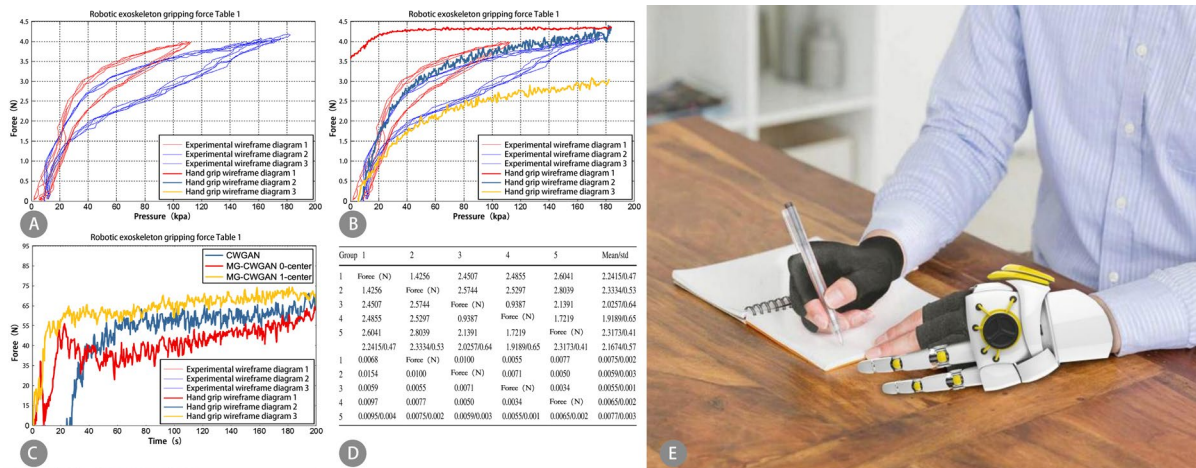


Figure 5. wireframe diagram test integration of the HCI-based robotic circuit drive system with snapshots in performing daily activity tasks. (A) Experiment 1 yields a linear force map of the maximum force on the human hand without the robotic exoskeleton. (B) Plot of the applied pressure (kpa) values for the manipulation experiment showing daily behaviors and activities. (C) The manipulators are all capable of simply grasping/grasping/lifting objects of different shapes in horizontal phase. (D) Summary data by integrating the first three experiments for the first three experiments of 9 robotic data. (E) Men's daily writing experiment test.

VI. EXPERIMENTAL RESULTS OF MANIPULATOR WITH HUMAN-COMPUTER INTERACTION

In this study, the maximum range of motion tolerance and output force of the circuit-driven system were tested, while demonstrating the operation of daily behaviors and activities. As shown in Figure 5 below, the researchers set the value of pressure to kpa by setting the value range to the maximum acceptable range of 180kpa for the everyday human hand; and set the minimum value to the initial force of 0kpa for the hand (the minimum value is to eliminate the factor under the influence of environmental gravity). Experiment 1 yielded the human hand without wearing a robot maximum linear force of 114kpa, while Force (N) value of 4.1; experiment 2 can be derived from the human arm withstand force of 182kpa, while Force (N) value of 4.3; experiment 3 data can be derived from the human arm withstand force of 173kpa, while Force (N) value of 4.1 (Figure 5A). Through experiments wearing robotic grip wireframe 1, can be derived from the human hand wearing robotic maximum linear force of 183kpa, while Force (N)

value of 4.4; wireframe 2 data can be derived from the robotic withstand force 182kpa, while Force (N) value of 4.4; wireframe 3 data can be derived from the robotic withstand force 185kpa, while Force (N) value of 3.1. The results of the six experiments conclude that the maximum range of motion withstood by the robot circuit drive system is similar to the parameters of the force data of the output force on the human hand, while showing the value of the pressure(kpa) imposed by the operation experiment of daily behavior and activities (Figure 5B). The study set the horizontal axis as time/second and the vertical axis as Force(N). The first experimental phase time was 200 seconds, while the applied force began to enter the peak point after 20 seconds, when the test reached the maximum set value of 200 seconds time, the test results robot Force(N) test values in the range of 55-65N. The second experimental stage time of 200 seconds, while the applied force began to enter the peak point after 60 seconds, the robot test value after 120 seconds Force (N) value of 40-55N range. When the test reached the maximum set value of 200 seconds time,

we found that the robot Forcee(N) test value is within the range of 55-65N. The third stage was similar to the first stage test results, and the manipulators were all able to simply grasp/grip/lift objects of different shapes under the horizontal stage (Figure 5C). By integrating the summary data from the first three experiments for the first three experiments 9 robotic data (Figure 5D). A snapshot of the HCI-based robotic circuit-driven system in performing daily activity tasks was tested, including men's daily writing experiments tested to better go about operating to enhance the quality of life of patients. (Figure 5E).

VII. CONCLUSION AND FUTURE WORK

This research team through the verification of human-computer interaction based robotic circuit drive system in the medical practicality, through from prototype design, prototype development test, structural design test, simulation of real kinematic analysis diagram, circuit drive system simulation and test will be developed by the product research are integrated into the robotic mechatronics, the shortcomings of this paper is the lack of sufficient research software and hardware matching degree, after continuous testing, design development and control system test can feedback good results.

One of the most innovative programs and future industries, this research focuses on the design and development of the robot for human-computer interaction, solving the need for a large number of nursing resources also with human resources in society. The development of a more skin-friendly, safer, and less expensive human-computer interaction-based robotic system (HES) that can operate by itself is sufficient to support all daily living activities (ADLs) of the hand. Finally, the physical field study experiments also show that the device is flexible enough for anyone to customize a suitable solution for use. Future design work will further improve the integration of the system, including the development of 3D customization of the hand system and synchronization of the left and right gloves for assisted use operations, including household chores, assistance with knives and forks, etc.

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